📊 Day 12: Semantic Matching Engine

Objective

To measure semantic similarity between resumes and job descriptions.

Approach

- Convert text into TF-IDF vectors
- Compute cosine similarity between resume and JD
- Generate similarity score

Features

- Captures contextual similarity
- Works beyond keyword matching
- Lightweight and fast

Accuracy

- Similar roles: 75–90%
- Different domains: 50–70%

Limitations

- Not as powerful as deep learning models
- Sensitive to text quality

Improvements

- Use Sentence Transformers (BERT)
- Fine-tune embeddings
- Add domain-specific training data